

INCIDENT RESPONSE & DIGITAL FORENSICS PLAYBOOK

Week 3: Containment, Eradication, and Recovery

Post-Phishing Endpoint Response and Recovery Documentation

Field	Value
Incident Type	Phishing / Credential-Harvesting Simulation
Target Endpoint	Windows 10 Workstation
Victim IP Address	192.168.92.133
Phishing Server	Kali Linux / GoPhish
Phishing Server IP	192.168.92.128
Primary Attack Vector	Email phishing link leading to a fake Secure Document Portal
Week 3 Status	Contained, eradicated, and recovered

Prepared for Project 5 - Incident Response and Digital Forensics Playbook

Scope Note: This report documents an authorized, isolated lab simulation. No production systems were targeted.

1. Document Purpose and Scope

This report documents Week 3 of the incident response lifecycle for a phishing simulation performed in an isolated virtual laboratory. Week 2 established that the victim endpoint received a phishing email, opened the message, accessed a fake Secure Document Portal hosted on the Kali/GoPhish server, and submitted credentials over HTTP. Week 3 focuses on responding to that confirmed simulated incident by containing the affected endpoint, removing phishing-related artifacts, checking for persistence indicators, and safely restoring network operation.

The scope of this report is limited to the post-phishing endpoint response workflow. The report does not claim that a full enterprise forensic investigation was performed. Instead, it provides a clear and defensible response record using the available lab evidence: firewall controls, network isolation, process review, startup review, scheduled task review, active connection review, browser artifact cleanup, and recovery validation.

2. Executive Summary

Following the Week 2 phishing simulation, the Windows 10 victim endpoint was treated as potentially exposed because the user interacted with a deceptive email link and submitted credentials to a fake login page hosted on the GoPhish phishing server. The immediate objective of Week 3 was to prevent further communication with the phishing server, preserve operational control of the endpoint, remove browser-side remnants of the phishing interaction, and validate that the system could be safely returned to network operation while keeping the phishing server blocked.

Containment was achieved using a host-based Windows Defender Firewall outbound blocking rule targeting 192.168.92.128, followed by validation using ping and PowerShell Test-NetConnection. A hard isolation action was then performed by disabling the Ethernet0 adapter. Eradication actions included browser data cleanup, process review, startup item review, scheduled task review, and active network connection inspection. Recovery was completed by re-enabling Ethernet0, confirming that the victim endpoint regained 192.168.92.133, validating normal connectivity to the default gateway 192.168.92.2, and confirming that HTTP access to 192.168.92.128:80 remained blocked.

Phase	Outcome
Containment	Firewall rule created; traffic to 192.168.92.128 blocked; endpoint isolated by disabling Ethernet0.
Eradication	Browser data cleared; processes, startup entries, scheduled tasks, and active connections reviewed.
Recovery	Ethernet0 re-enabled; IP configuration restored; gateway communication verified.
Final Validation	The endpoint regained normal network operation, while access to 192.168.92.128:80 remained blocked.

3. Lab Environment

The environment consisted of a virtual Windows 10 endpoint and a Kali Linux machine hosting the GoPhish phishing infrastructure. The lab network used the 192.168.92.0/24 address range. The Windows endpoint served as the victim machine, while Kali operated as the attacker-controlled phishing server used for simulation and evidence generation.

Asset	Role	IP Address	Purpose
Windows 10 Workstation	Victim endpoint	192.168.92.133	System that received the email, clicked the link, and submitted credentials during Week 2.
Kali Linux	GoPhish / phishing server	192.168.92.128	Hosted the fake Secure Document Portal and received HTTP traffic during Week 2.
Default Gateway	Lab network gateway	192.168.92.2	Used during recovery validation to confirm restored network connectivity.

4. Root Cause Analysis

The root cause was successful user interaction with a deceptive phishing email. The email impersonated a secure document-sharing workflow and directed the victim to a fake Secure Document Portal hosted on the GoPhish server. The user then submitted credentials to the phishing page, creating a simulated credential exposure event.

The technical contributing factor was the absence of inline URL filtering or browser-level blocking for the lab phishing URL. The human contributing factor was trust in a convincing document-verification message. Because the simulated credential submission occurred over HTTP, the incident also demonstrated the risk of cleartext credential capture in unencrypted web traffic.

RCA Item	Finding
Initial Vector	Phishing email titled Shared Document Requires Verification.
User Action	The victim opened the email, clicked the link, and submitted credentials.
Infrastructure	Fake Secure Document Portal hosted on Kali/GoPhish at 192.168.92.128.
Primary Root Cause	User trusted a deceptive email link and interacted with the fake portal.
Technical Gap	No URL filtering or phishing-site blocking prevented the browser from reaching the lab phishing page.
Simulated Impact	Credential exposure risk and confirmed outbound HTTP communication from the victim endpoint to the phishing server.

5. Containment Phase

The containment phase focused on stopping communication between the Windows victim endpoint and the GoPhish phishing server. A layered approach was used: first, a Windows Defender Firewall outbound block rule was created for the phishing server IP address; second, network connectivity was validated as blocked; third, the endpoint network adapter was disabled to provide hard isolation from the lab network.

5.1. Host-Based Firewall Blocking

A host-based outbound rule was created on the Windows endpoint using PowerShell. The rule was named Block Gophish Phishing Server and targeted the remote IP address 192.168.92.128. The rule was configured with Direction = Outbound and Action = Block, which prevented the victim endpoint from initiating new outbound communication to the phishing server.

Item	Details
Command	New-NetFirewallRule -DisplayName "Block Gophish Phishing Server" -Direction Outbound -Action Block -RemoteAddress 192.168.92.128
Validation Command	Get-NetFirewallRule -DisplayName "Block Gophish Phishing Server"
Rule State	Enabled = True; Action = Block; Direction = Outbound; Profile = Any

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Block Gophish Phishing Server" -Direction Outbound -Action Block -RemoteAddress 192.168.92.128

Name                : {42be0d1e-10cd-4a18-86f7-73bd5d22a2af}
DisplayName          : Block Gophish Phishing Server
Description          :
DisplayGroup        :
Group               :
Enabled             : True
Profile             : Any
Platform           : {}
Direction          : Outbound
Action             : Block
EdgeTraversalPolicy : Block
LooseSourceMapping  : False
LocalOnlyMapping    : False
Owner              :
PrimaryStatus       : OK
Status             : The rule was parsed successfully from the store. (65536)
EnforcementStatus   : NotApplicable
PolicyStoreSource   : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PolicyAppId        :

PS C:\Windows\system32> Get-NetFirewallRule -DisplayName "Block Gophish Phishing Server"

Name                : {42be0d1e-10cd-4a18-86f7-73bd5d22a2af}
DisplayName          : Block Gophish Phishing Server
Description          :
DisplayGroup        :
Group               :
Enabled             : True
Profile             : Any
Platform           : {}
Direction          : Outbound
Action             : Block
EdgeTraversalPolicy : Block
LooseSourceMapping  : False
LocalOnlyMapping    : False
Owner              :
PrimaryStatus       : OK
Status             : The rule was parsed successfully from the store. (65536)
EnforcementStatus   : NotApplicable
PolicyStoreSource   : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PolicyAppId        :
```

Figure 1: PowerShell creation and verification of the outbound firewall rule blocking the GoPhish phishing server.

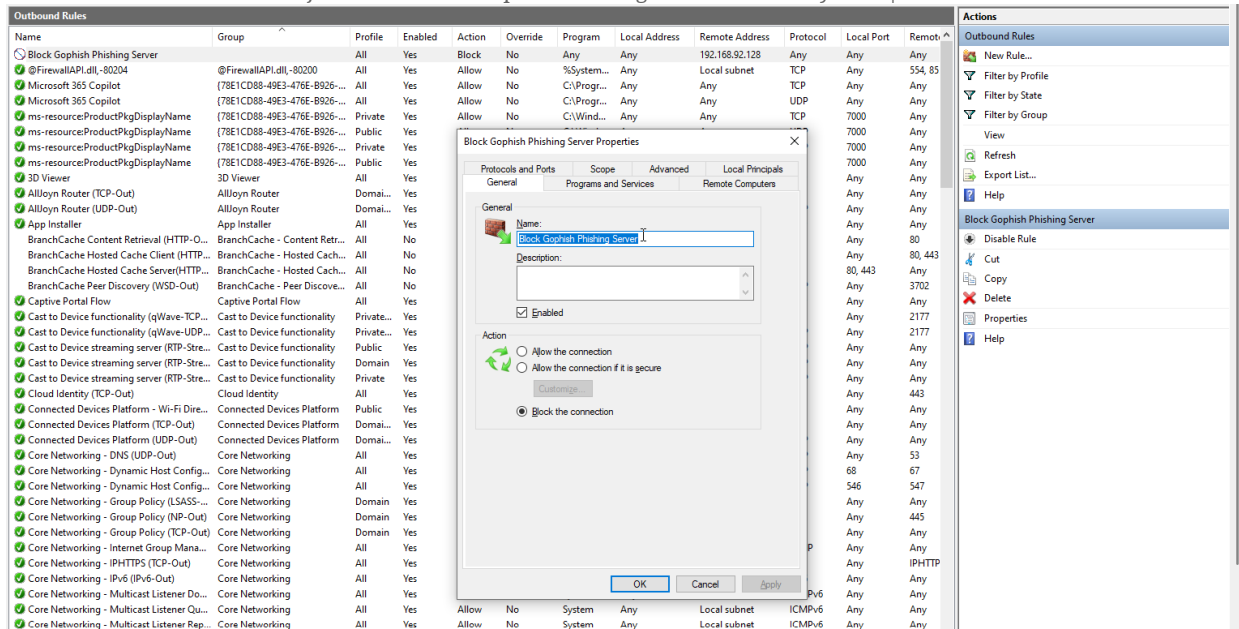


Figure 2: Windows Defender Firewall GUI confirmation that the outbound rule is enabled, set to Block, and scoped to 192.168.92.128.

5.2. Connectivity Blocking Validation

After the firewall rule was applied, connectivity was tested using ping and Test-NetConnection. The ping test returned General failure and 100% packet loss. The PowerShell TCP test against port 80 also failed, confirming that the victim endpoint could no longer reach the phishing web service.

Validation Test	Observed Result
ping 192.168.92.128	Failed with 100% packet loss.
Test-NetConnection 192.168.92.128 -Port 80	TcpTestSucceeded = False; PingSucceeded = False.

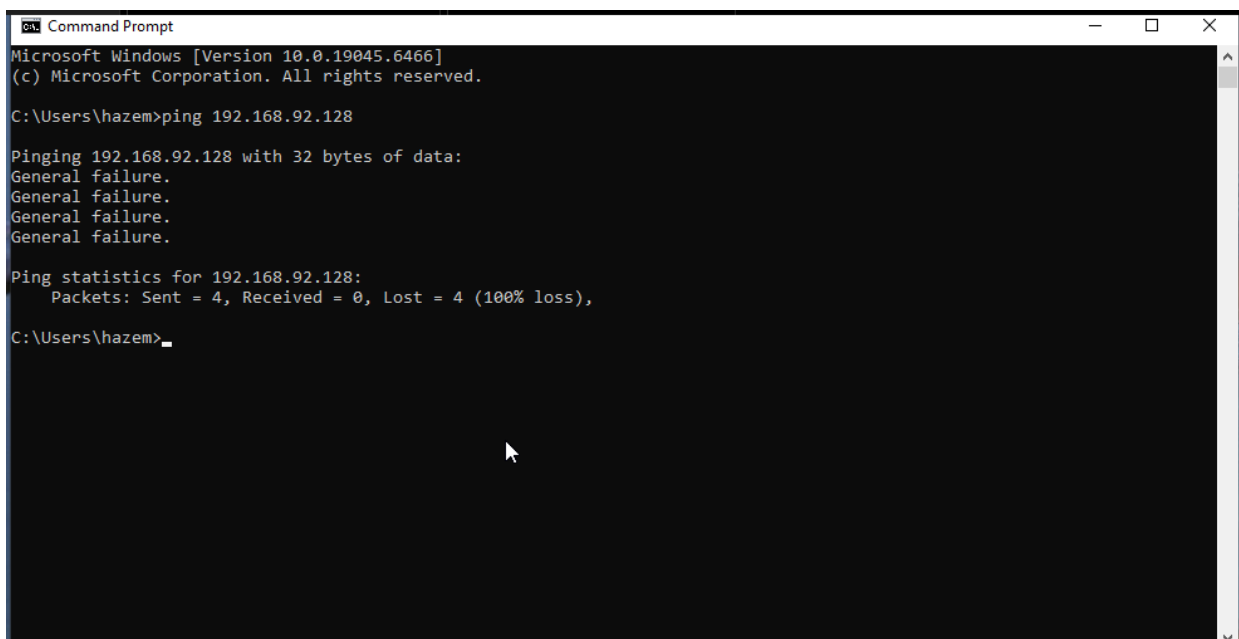
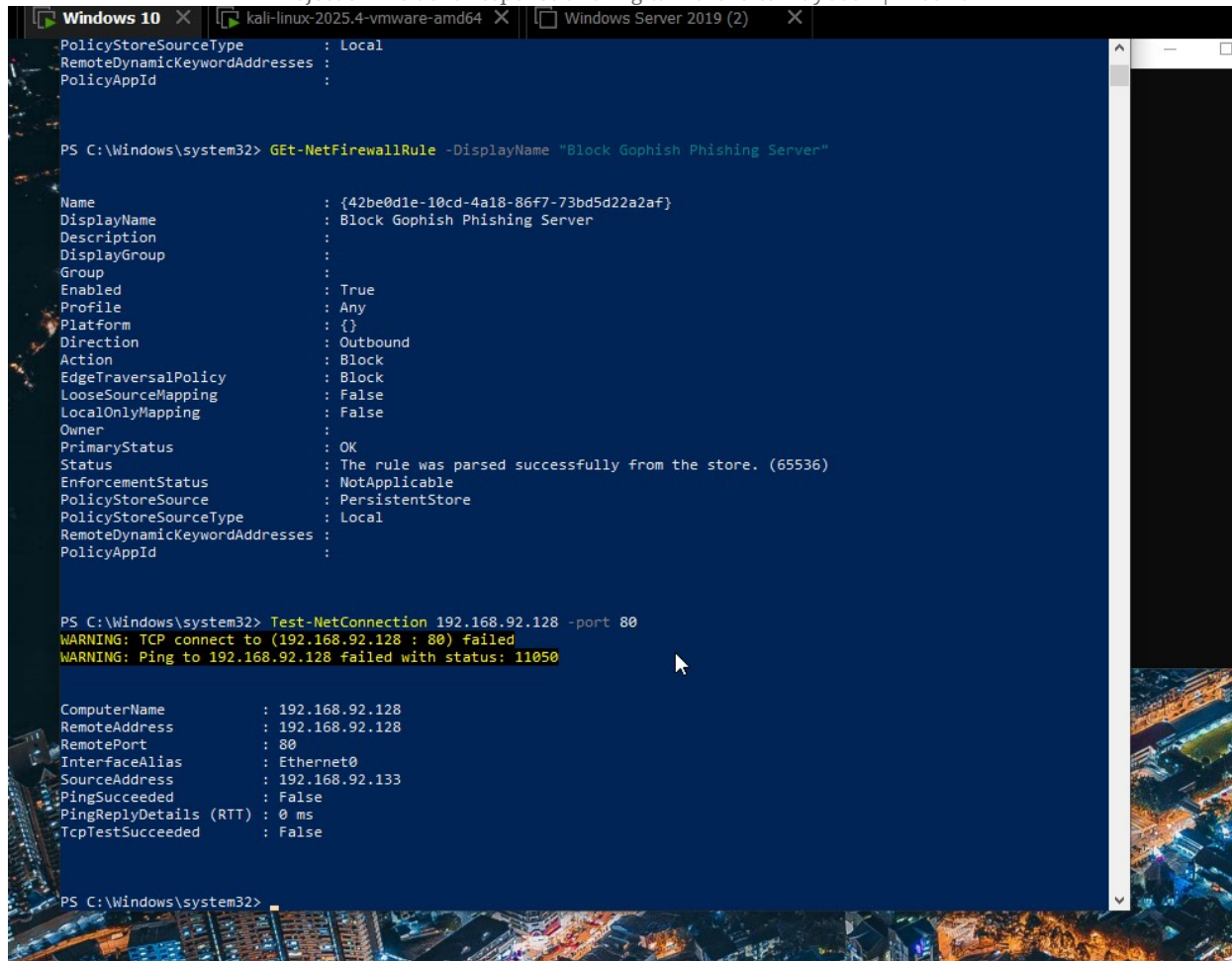


Figure 3: Post-containment ping test showing 100% packet loss to the phishing server 192.168.92.128.



```

Windows 10 X kali-linux-2025.4-vmware-amd64 X Windows Server 2019 (2) X
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PolicyAppId :

PS C:\Windows\system32> Get-NetFirewallRule -DisplayName "Block Gophish Phishing Server"

Name                           : {42be0d1e-10cd-4a18-86f7-73bd5d22a2af}
DisplayName                     : Block Gophish Phishing Server
Description                     :
DisplayGroup                    :
Group                           :
Enabled                         : True
Profile                         : Any
Platform                       : {}
Direction                      : Outbound
Action                         : Block
EdgeTraversalPolicy             : Block
LooseSourceMapping              : False
LocalOnlyMapping                : False
Owner                           :
PrimaryStatus                   : OK
Status                         : The rule was parsed successfully from the store. (65536)
EnforcementStatus               : NotApplicable
PolicyStoreSource               : PersistentStore
PolicyStoreSourceType           : Local
RemoteDynamicKeywordAddresses :
PolicyAppId                     :

PS C:\Windows\system32> Test-NetConnection 192.168.92.128 -port 80
WARNING: TCP connect to (192.168.92.128 : 80) failed
WARNING: Ping to 192.168.92.128 failed with status: 11050

ComputerName      : 192.168.92.128
RemoteAddress     : 192.168.92.128
RemotePort        : 80
InterfaceAlias    : Ethernet0
SourceAddress     : 192.168.92.133
PingSucceeded     : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False

PS C:\Windows\system32>

```

Figure 4: PowerShell Test-NetConnection showing that TCP connectivity to 192.168.92.128:80 failed after containment.

5.3. Endpoint Network Isolation

After traffic blocking was validated, a hard isolation step was performed by disabling the Ethernet0 adapter from Network Connections. This reduced the risk of any further network communication while eradication and verification checks were performed.

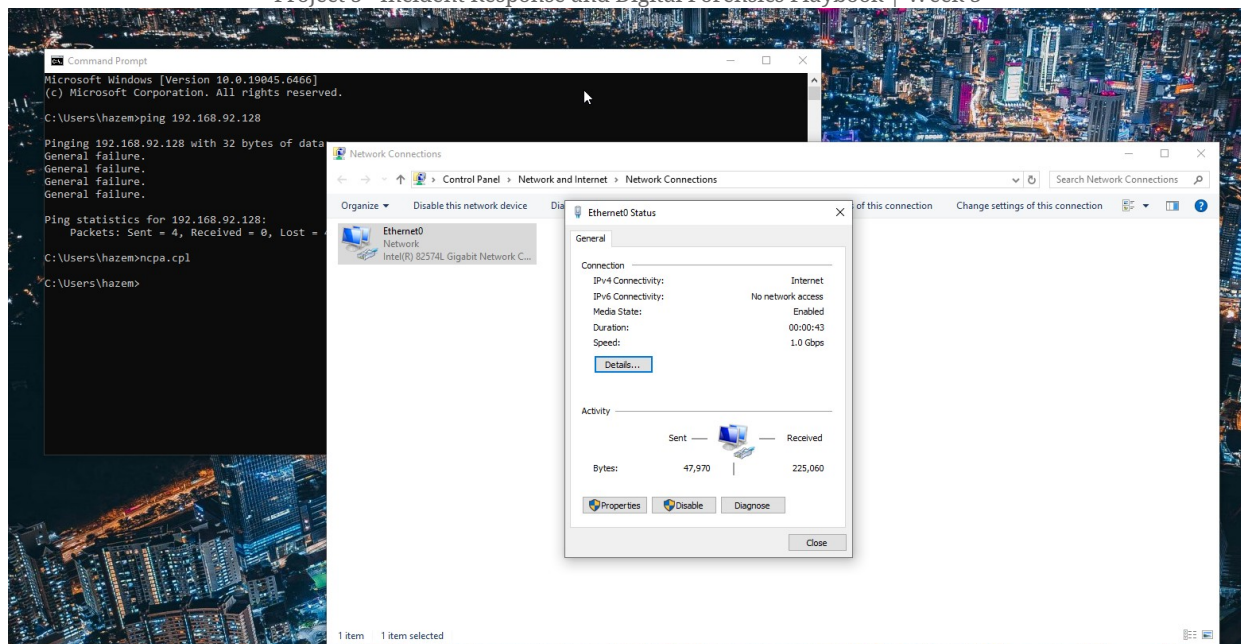


Figure 5: Network adapter status before full isolation, confirming the adapter was still enabled before the hard-disable action.

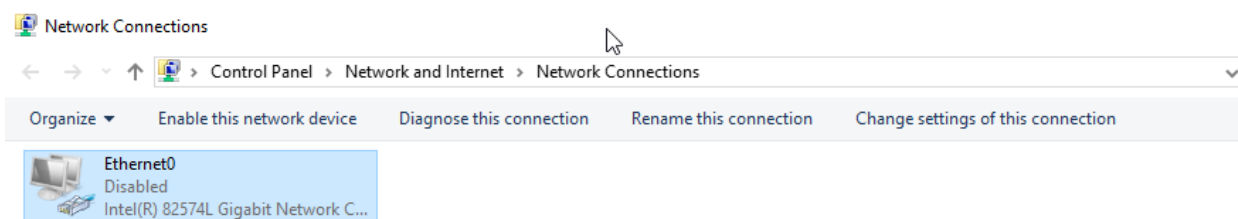


Figure 6: Ethernet0 disabled from Network Connections to fully isolate the Windows endpoint from the lab network.

6. Eradication and Verification Phase

The eradication phase focused on removing phishing-related browser artifacts and reviewing common persistence locations. Because this was a phishing simulation rather than a malware execution

scenario, the wording is deliberately conservative: the checks did not identify suspicious persistence indicators related to the phishing simulation. This does not claim that a full malware reverse-engineering or disk-forensic examination was performed.

6.1. Browser Artifact Cleanup

Microsoft Edge browsing data was cleared to remove cached content, cookies, and browsing traces associated with the fake Secure Document Portal. This step is appropriate in credential-harvesting scenarios because the browser may store session artifacts, cached page content, or local remnants of the phishing page.

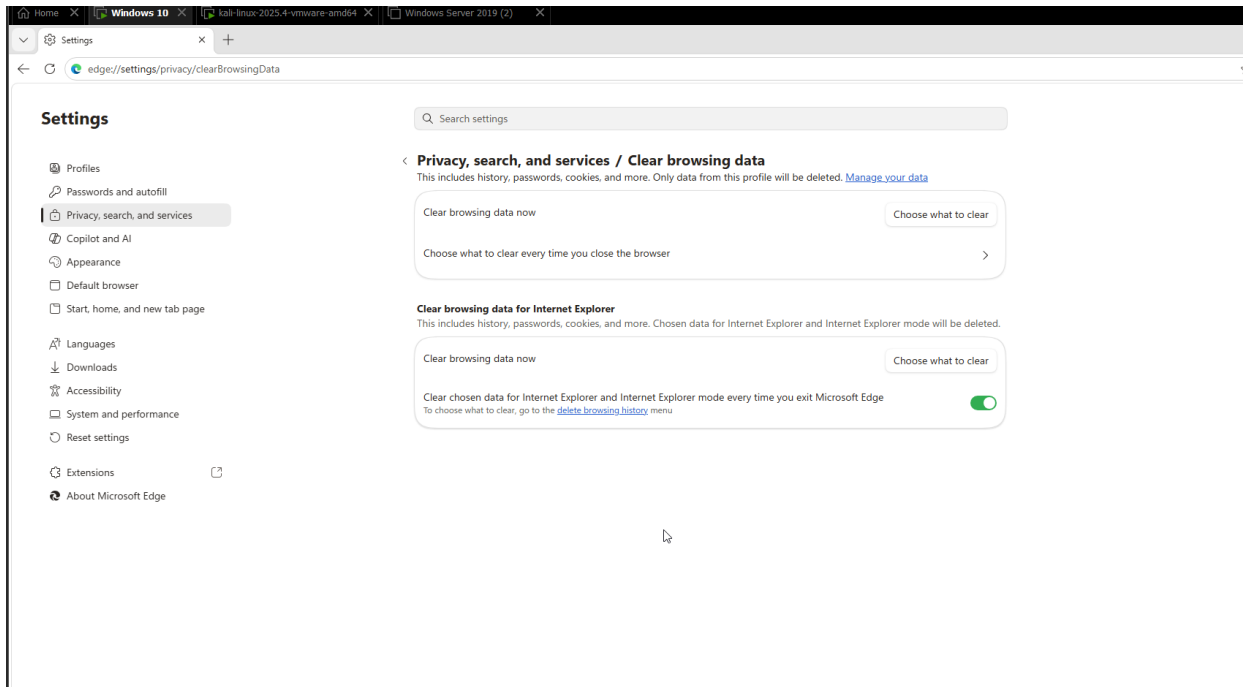


Figure 7: Microsoft Edge browsing data cleanup page after removing browser artifacts associated with the phishing interaction.

6.2. Active Process Review

The tasklist command was used to review active processes on the victim endpoint. The visible output showed standard Windows services and browser-related processes. No process in the provided screenshots was identified as directly associated with the phishing simulation beyond normal browser activity.

```
C:\Users\hazem>wmic startup get caption,command
Caption                                Command
OneDrive                              "C:\Users\hazem\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background
MicrosoftEdgeAutoLaunch_3358D57C7D1E8609EFF876EE54E1F4EC "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --no-startup-window --win-session-start
SecurityHealth                         %windir%\system32\SecurityHealthSystray.exe

C:\Users\hazem>
```

Figure 8: Initial tasklist output used to review active processes on the Windows victim endpoint.

Process Name	PID	Session Name	Session ID	Private Bytes
svchost.exe	7060	Console	1	26,328 K
SearchApp.exe	7380	Console	1	105,192 K
RuntimeBroker.exe	7476	Console	1	46,044 K
RuntimeBroker.exe	7708	Console	1	26,992 K
SecurityHealthSystray.exe	1308	Console	1	8,572 K
SecurityHealthService.exe	372	Services	0	14,616 K
OneDrive.exe	8004	Console	1	85,580 K
OneDrive.Sync.Service.exe	8584	Console	1	63,984 K
msedge.exe	8976	Console	1	219,836 K
msedge.exe	9164	Console	1	8,700 K
msedge.exe	8996	Console	1	52,552 K
msedge.exe	8988	Console	1	41,676 K
msedge.exe	8964	Console	1	17,748 K
elevation_service.exe	8492	Services	0	10,636 K
svchost.exe	9840	Services	0	6,760 K
M365Copilot.exe	6632	Console	1	83,788 K
RuntimeBroker.exe	9660	Console	1	14,496 K
msedgebview2.exe	9780	Console	1	14,604 K
msedgebview2.exe	9892	Console	1	8,552 K
msedgebview2.exe	10132	Console	1	9,132 K
msedgebview2.exe	8484	Console	1	2,932 K
msedgebview2.exe	8480	Console	1	1,552 K
TextInputHost.exe	9028	Console	1	37,152 K
msedgebview2.exe	8396	Console	1	1,808 K
dllhost.exe	10544	Console	1	12,156 K
svchost.exe	9808	Services	0	18,852 K
msedge.exe	10140	Console	1	41,564 K
msedge.exe	6712	Console	1	23,816 K
FileCoAuth.exe	5676	Console	1	30,772 K
SearchApp.exe	2424	Console	1	51,880 K
msedgebview2.exe	2880	Console	1	83,416 K
msedgebview2.exe	2808	Console	1	6,604 K
msedgebview2.exe	8520	Console	1	37,340 K
msedgebview2.exe	9676	Console	1	30,828 K
msedgebview2.exe	8512	Console	1	16,340 K
msedgebview2.exe	2236	Console	1	52,572 K
UserOOBEBroker.exe	2152	Console	1	9,512 K
mmc.exe	7904	Console	1	13,692 K
svchost.exe	2404	Services	0	15,476 K
powershell.exe	6096	Console	1	147,420 K
conhost.exe	3276	Console	1	19,440 K
WmiPrvSE.exe	11088	Services	0	15,488 K
cmd.exe	4528	Console	1	15,700 K
conhost.exe	2660	Console	1	22,264 K
svchost.exe	7968	Services	0	11,740 K
explorer.exe	5892	Console	1	54,232 K
svchost.exe	11156	Services	0	13,904 K
dllhost.exe	10864	Console	1	8,228 K
svchost.exe	4188	Services	0	7,616 K
svchost.exe	6292	Services	0	7,864 K
svchost.exe	4516	Console	1	13,696 K
svchost.exe	4088	Services	0	10,912 K
svchost.exe	10612	Services	0	24,606 K
msedge.exe	7196	Console	1	115,892 K
msedge.exe	7400	Console	1	31,640 K
tasklist.exe	7144	Console	1	9,352 K

Figure 9: Continued tasklist output showing user-session and browser-related processes during endpoint process review.

6.3. Startup Item Review

Startup entries were reviewed using `wmic startup get caption,command`. The visible entries included OneDrive, Microsoft Edge auto-launch, and Windows Security notification components. No suspicious startup entry related to the phishing simulation was identified in the visible output.

```
C:\Users\hazem>tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	92 K
Registry	92	Services	0	28,244 K
smss.exe	316	Services	0	916 K
csrss.exe	432	Services	0	4,860 K
wininit.exe	508	Services	0	5,920 K
csrss.exe	516	Console	1	5,184 K
services.exe	588	Services	0	8,316 K
winlogon.exe	616	Console	1	9,340 K
lsass.exe	640	Services	0	20,836 K
svchost.exe	772	Services	0	28,484 K
fontdrvhost.exe	780	Services	0	3,516 K
fontdrvhost.exe	788	Console	1	4,552 K
svchost.exe	888	Services	0	15,240 K
svchost.exe	944	Services	0	9,240 K
svchost.exe	368	Services	0	9,004 K
svchost.exe	380	Services	0	11,308 K
dwm.exe	420	Console	1	113,096 K
svchost.exe	1052	Services	0	37,036 K
svchost.exe	1108	Services	0	8,076 K
svchost.exe	1160	Services	0	5,420 K
svchost.exe	1200	Services	0	6,848 K
svchost.exe	1220	Services	0	12,580 K
svchost.exe	1240	Services	0	6,784 K
svchost.exe	1268	Services	0	12,972 K
svchost.exe	1300	Services	0	5,256 K
Memory Compression	1428	Services	0	191,612 K
svchost.exe	1464	Services	0	7,680 K
svchost.exe	1488	Services	0	14,044 K
svchost.exe	1524	Services	0	6,136 K
svchost.exe	1576	Services	0	10,960 K
svchost.exe	1584	Services	0	6,464 K
svchost.exe	1636	Services	0	7,160 K
svchost.exe	1828	Services	0	11,276 K
svchost.exe	1856	Services	0	9,412 K
svchost.exe	1920	Services	0	7,304 K
svchost.exe	1928	Services	0	5,796 K
svchost.exe	1936	Services	0	8,464 K
svchost.exe	720	Services	0	11,676 K
svchost.exe	1156	Services	0	9,120 K
svchost.exe	456	Services	0	17,084 K
svchost.exe	2204	Services	0	6,964 K
spoolsv.exe	2220	Services	0	12,596 K
svchost.exe	2316	Services	0	16,752 K
svchost.exe	2332	Services	0	7,084 K
svchost.exe	2596	Services	0	12,680 K
svchost.exe	2608	Services	0	47,168 K
svchost.exe	2616	Services	0	33,804 K
svchost.exe	2636	Services	0	17,680 K
MpDefenderCoreService.exe	2696	Services	0	16,796 K

Figure 10: Startup entries reviewed using `wmic startup get caption,command`.

6.4. Scheduled Task Review

Scheduled tasks were reviewed using `schtasks /query`. The output showed standard Microsoft, OneDrive, .NET Framework, and Windows application experience tasks. No unauthorized scheduled task directly related to the phishing simulation was identified in the visible evidence.

```

C:\Users\hazem>schtasks /query

Folder: \
TaskName
Next Run Time
Status
-----
OneDrive Reporting Task-S-1-5-21-6486437 6/27/2026 10:42:40 PM Ready
OneDrive Standalone Update Task-S-1-5-21 6/28/2026 12:16:16 AM Ready
OneDrive Startup Task-S-1-5-21-648643775 N/A Ready

Folder: \Microsoft
TaskName
Next Run Time
Status
-----
INFO: There are no scheduled tasks presently available at your access level.

Folder: \Microsoft\OneCore
TaskName
Next Run Time
Status
-----
INFO: There are no scheduled tasks presently available at your access level.

Folder: \Microsoft\Windows
TaskName
Next Run Time
Status
-----
INFO: There are no scheduled tasks presently available at your access level.

Folder: \Microsoft\Windows\ .NET Framework
TaskName
Next Run Time
Status
-----
.NET Framework NGEN v4.0.30319 N/A Ready
.NET Framework NGEN v4.0.30319 64 N/A Ready
.NET Framework NGEN v4.0.30319 64 Critic N/A Disabled
.NET Framework NGEN v4.0.30319 Critical N/A Disabled

Folder: \Microsoft\Windows\Active Directory Rights Management Services Client
TaskName
Next Run Time
Status
-----
AD RMS Rights Policy Template Management N/A Disabled
AD RMS Rights Policy Template Management N/A Ready

Folder: \Microsoft\Windows\AppID
TaskName
Next Run Time
Status
-----
PolicyConverter N/A Disabled
VerifiedPublisherCertStoreCheck N/A Disabled

Folder: \Microsoft\Windows\Application Experience
TaskName
Next Run Time
Status
-----
MareBackup N/A Ready
Microsoft Compatibility Appraiser 6/28/2026 3:11:24 AM Ready
PcaPatchDbTask 6/27/2026 3:11:20 PM Ready
PcaWallpaperAppDetect 6/28/2026 12:00:32 PM Ready
ProgramDataUpdater N/A Ready
StartupAppTask N/A Ready

```

Figure 11: Scheduled task review using `schtasks /query` to check for unauthorized scheduled persistence.

6.5. Active Network Connection Review

Active network connections were reviewed using `netstat -ano`. The visible output primarily showed local listening ports and loopback connections such as 127.0.0.1. No active connection to the phishing server 192.168.92.128 was visible in the submitted `netstat` screenshots after containment and cleanup.

```

C:\Users\hazem>netstat -ano

Active Connections

Proto Local Address          Foreign Address         State       PID
TCP   0.0.0.0:135             0.0.0.0:0               LISTENING   888
TCP   0.0.0.0:445             0.0.0.0:0               LISTENING   4
TCP   0.0.0.0:5040            0.0.0.0:0               LISTENING   5972
TCP   0.0.0.0:7680            0.0.0.0:0               LISTENING   3584
TCP   0.0.0.0:8000            0.0.0.0:0               LISTENING   2940
TCP   0.0.0.0:8089            0.0.0.0:0               LISTENING   2940
TCP   0.0.0.0:8191            0.0.0.0:0               LISTENING   4736
TCP   0.0.0.0:49664           0.0.0.0:0               LISTENING   640
TCP   0.0.0.0:49665           0.0.0.0:0               LISTENING   508
TCP   0.0.0.0:49666           0.0.0.0:0               LISTENING   1052
TCP   0.0.0.0:49667           0.0.0.0:0               LISTENING   1488
TCP   0.0.0.0:49668           0.0.0.0:0               LISTENING   2220
TCP   0.0.0.0:49671           0.0.0.0:0               LISTENING   588
TCP   127.0.0.1:8065          0.0.0.0:0               LISTENING   3708
TCP   127.0.0.1:8089          127.0.0.1:52364         TIME_WAIT   0
TCP   127.0.0.1:8089          127.0.0.1:52365         TIME_WAIT   0
TCP   127.0.0.1:8089          127.0.0.1:52378         ESTABLISHED 2940
TCP   127.0.0.1:8089          127.0.0.1:52390         ESTABLISHED 2940
TCP   127.0.0.1:8089          127.0.0.1:52391         ESTABLISHED 2940
TCP   127.0.0.1:8191          127.0.0.1:49713         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49715         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49716         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49717         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49718         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49719         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49721         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49723         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49724         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49725         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49727         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49729         ESTABLISHED 4736
TCP   127.0.0.1:8191          127.0.0.1:49731         ESTABLISHED 4736
TCP   127.0.0.1:8194          0.0.0.0:0               LISTENING   5180
TCP   127.0.0.1:8194          127.0.0.1:49740         ESTABLISHED 5180
TCP   127.0.0.1:8194          127.0.0.1:49744         ESTABLISHED 5180
TCP   127.0.0.1:49697         0.0.0.0:0               LISTENING   4396
TCP   127.0.0.1:49697         127.0.0.1:49807         ESTABLISHED 4396
TCP   127.0.0.1:49697         127.0.0.1:49813         ESTABLISHED 4396
TCP   127.0.0.1:49697         127.0.0.1:49814         ESTABLISHED 4396
TCP   127.0.0.1:49697         127.0.0.1:49815         ESTABLISHED 4396
TCP   127.0.0.1:49697         127.0.0.1:49816         ESTABLISHED 4396
TCP   127.0.0.1:49697         127.0.0.1:49818         ESTABLISHED 4396
TCP   127.0.0.1:49698         0.0.0.0:0               LISTENING   4396
TCP   127.0.0.1:49713         127.0.0.1:8191         ESTABLISHED 5668
TCP   127.0.0.1:49715         127.0.0.1:8191         ESTABLISHED 5668
TCP   127.0.0.1:49716         127.0.0.1:8191         ESTABLISHED 5668
TCP   127.0.0.1:49717         127.0.0.1:8191         ESTABLISHED 5668
TCP   127.0.0.1:49718         127.0.0.1:8191         ESTABLISHED 5668

```

Figure 12: Active connection review using netstat -ano, first output section.

Command Prompt				
TCP	127.0.0.1:49721	127.0.0.1:8191	ESTABLISHED	5668
TCP	127.0.0.1:49723	127.0.0.1:8191	ESTABLISHED	2940
TCP	127.0.0.1:49724	127.0.0.1:8191	ESTABLISHED	2940
TCP	127.0.0.1:49725	127.0.0.1:8191	ESTABLISHED	2940
TCP	127.0.0.1:49727	127.0.0.1:8191	ESTABLISHED	2940
TCP	127.0.0.1:49728	0.0.0.0:0	LISTENING	5180
TCP	127.0.0.1:49728	127.0.0.1:49737	ESTABLISHED	5180
TCP	127.0.0.1:49729	127.0.0.1:8191	ESTABLISHED	2940
TCP	127.0.0.1:49731	127.0.0.1:8191	ESTABLISHED	2940
TCP	127.0.0.1:49737	127.0.0.1:49728	ESTABLISHED	4396
TCP	127.0.0.1:49740	127.0.0.1:8194	ESTABLISHED	4396
TCP	127.0.0.1:49744	127.0.0.1:8194	ESTABLISHED	4396
TCP	127.0.0.1:49753	0.0.0.0:0	LISTENING	5580
TCP	127.0.0.1:49754	0.0.0.0:0	LISTENING	5580
TCP	127.0.0.1:49754	127.0.0.1:49772	ESTABLISHED	5580
TCP	127.0.0.1:49763	0.0.0.0:0	LISTENING	5684
TCP	127.0.0.1:49764	0.0.0.0:0	LISTENING	5684
TCP	127.0.0.1:49764	127.0.0.1:49768	ESTABLISHED	5684
TCP	127.0.0.1:49768	127.0.0.1:49764	ESTABLISHED	4396
TCP	127.0.0.1:49772	127.0.0.1:49754	ESTABLISHED	4396
TCP	127.0.0.1:49792	0.0.0.0:0	LISTENING	5056
TCP	127.0.0.1:49792	127.0.0.1:49833	ESTABLISHED	5056
TCP	127.0.0.1:49794	0.0.0.0:0	LISTENING	5056
TCP	127.0.0.1:49794	127.0.0.1:49798	ESTABLISHED	5056
TCP	127.0.0.1:49798	127.0.0.1:49794	ESTABLISHED	4396
TCP	127.0.0.1:49807	127.0.0.1:49697	ESTABLISHED	5056
TCP	127.0.0.1:49813	127.0.0.1:49697	ESTABLISHED	5056
TCP	127.0.0.1:49814	127.0.0.1:49697	ESTABLISHED	5056
TCP	127.0.0.1:49815	127.0.0.1:49697	ESTABLISHED	5056
TCP	127.0.0.1:49816	127.0.0.1:49697	ESTABLISHED	5056
TCP	127.0.0.1:49818	127.0.0.1:49697	ESTABLISHED	5056
TCP	127.0.0.1:49833	127.0.0.1:49792	ESTABLISHED	5056
TCP	127.0.0.1:52378	127.0.0.1:8089	ESTABLISHED	5684
TCP	127.0.0.1:52390	127.0.0.1:8089	ESTABLISHED	5684
TCP	127.0.0.1:52391	127.0.0.1:8089	ESTABLISHED	5684
TCP	[::]:135	[::]:0	LISTENING	888
TCP	[::]:445	[::]:0	LISTENING	4
TCP	[::]:7680	[::]:0	LISTENING	3584
TCP	[::]:49664	[::]:0	LISTENING	640
TCP	[::]:49665	[::]:0	LISTENING	508
TCP	[::]:49666	[::]:0	LISTENING	1052
TCP	[::]:49667	[::]:0	LISTENING	1488
TCP	[::]:49668	[::]:0	LISTENING	2220
TCP	[::]:49671	[::]:0	LISTENING	588
TCP	[::1]:42050	[::]:0	LISTENING	8584
UDP	0.0.0.0:123	*:*		1248
UDP	0.0.0.0:5050	*:*		5972
UDP	0.0.0.0:52357	*:*		9676
UDP	127.0.0.1:1900	*:*		9840
UDP	127.0.0.1:58266	*:*		2868
UDP	127.0.0.1:64956	*:*		9840
UDP	[::]:123	*:*		1248
UDP	[::1]:1900	*:*		9840

Figure 13: Active connection review using netstat -ano, second output section.

7. Recovery Phase

Recovery was performed only after containment and eradication checks had been completed. The objective was not simply to reconnect the system, but to confirm that normal network operation could resume while maintaining the block against the phishing server.

7.1. Network Re-Enablement

Ethernet0 was re-enabled from Network Connections. This restored the endpoint to the lab network after the isolation step.

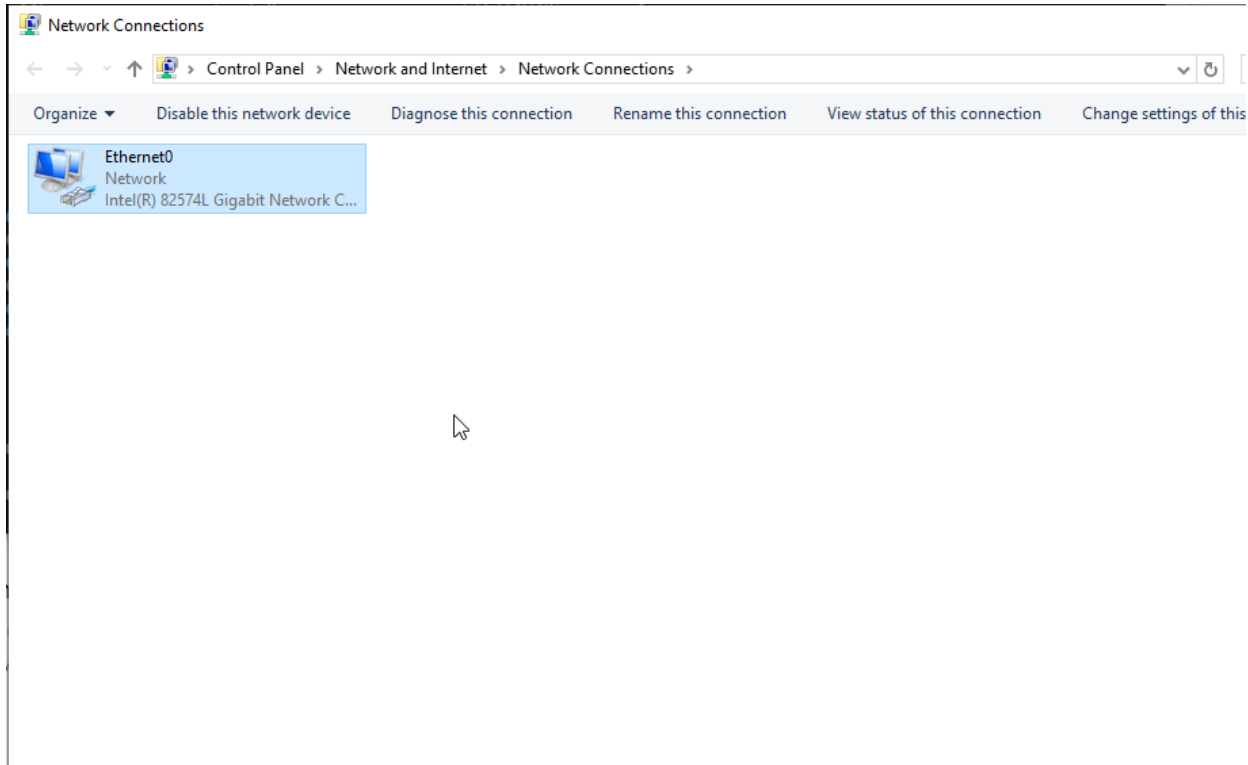


Figure 14: Ethernet0 re-enabled as part of the recovery phase.

7.2. Normal Connectivity Validation

After re-enabling the adapter, ipconfig confirmed that the endpoint regained the IPv4 address 192.168.92.133. Connectivity to the lab gateway 192.168.92.2 was successfully tested with ping, showing 0% packet loss. This confirmed that the endpoint had been restored to normal network operation inside the lab environment.

```
C:\ Command Prompt
C:\Users\hazem>ncpa.cpl

C:\Users\hazem>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::803c:8a9f:2020:d06c%2
    IPv4 Address. . . . . : 192.168.92.133
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.92.2

C:\Users\hazem>ping 192.168.92.2

Pinging 192.168.92.2 with 32 bytes of data:
Reply from 192.168.92.2: bytes=32 time<1ms TTL=128
Reply from 192.168.92.2: bytes=32 time<1ms TTL=128
Reply from 192.168.92.2: bytes=32 time<1ms TTL=128
Reply from 192.168.92.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.92.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\hazem>
```

Figure 15: Recovery validation showing IP restoration and successful communication with the default gateway 192.168.92.2.

7.3. Post-Recovery Blocking Validation

The final validation step retested TCP connectivity to the phishing server on port 80. The result remained `TcpTestSucceeded = False`. This is an important recovery control: the victim endpoint was operational again, but access to the known phishing server remained blocked.


```

PS C:\Windows\system32> Test-NetConnection 192.168.92.128 -port 80
WARNING: TCP connect to (192.168.92.128 : 80) failed
WARNING: Ping to 192.168.92.128 failed with status: 11050

ComputerName      : 192.168.92.128
RemoteAddress     : 192.168.92.128
RemotePort        : 80
InterfaceAlias    : Ethernet0
SourceAddress     : 192.168.92.133
PingSucceeded     : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False

PS C:\Windows\system32>

```

Figure 16: Final validation confirming that 192.168.92.128:80 remained blocked after recovery.

8. Incident Response Timeline

Time / Stage	Event
Week 2	Phishing email delivered, opened, and clicked; credentials submitted to the fake portal.
Week 2	Evidence collected from GoPhish, Wireshark, browser history, Sysmon, Splunk, and email headers.
Week 3 - Step 1	Created outbound Windows Defender Firewall rule to block 192.168.92.128.
Week 3 - Step 2	Validated containment using ping and Test-NetConnection to port 80.
Week 3 - Step 3	Disabled Ethernet0 for hard endpoint isolation.
Week 3 - Step 4	Cleared browser data to remove phishing-page artifacts.
Week 3 - Step 5	Reviewed active processes using tasklist.
Week 3 - Step 6	Reviewed startup entries using wmic startup.
Week 3 - Step 7	Reviewed scheduled tasks using schtasks /query.
Week 3 - Step 8	Reviewed active network connections using netstat -ano.
Week 3 - Step 9	Re-enabled Ethernet0 as part of recovery.
Week 3 - Step 10	Validated restored IP address and gateway connectivity.
Week 3 - Step 11	Confirmed that access to 192.168.92.128:80 remained blocked after recovery.

9. Containment, Eradication, and Recovery Action Matrix

Phase	Action	Tool / Command	Result
Containment	Created outbound firewall rule	New-NetFirewallRule	Traffic to 192.168.92.128 was blocked.
Containment	Validated blocking	ping; Test-NetConnection	Ping and TCP port 80 checks failed as expected.
Containment	Hard isolation	Network Connections / Ethernet0 disable	Endpoint was isolated from the lab network.
Eradication	Browser cleanup	Microsoft Edge settings	Browser-side phishing artifacts were removed.
Eradication	Process review	tasklist	No suspicious process directly related to the

			phishing simulation was identified in visible output.
Eradication	Startup review	wmic startup get caption,command	No suspicious startup entry related to the simulation was identified in visible output.
Eradication	Scheduled task review	schtasks /query	No unauthorized scheduled task directly related to the simulation was identified in visible output.
Eradication	Connection review	netstat -ano	No active connection to 192.168.92.128 was visible in the submitted output.
Recovery	Re-enabled network	Network Connections / Ethernet0 enable	Normal adapter status restored.
Recovery	Gateway validation	ipconfig; ping 192.168.92.2	Endpoint regained 192.168.92.133 and communicated with gateway.
Final validation	Retested phishing server block	Test-NetConnection 192.168.92.128 -Port 80	GoPhish server remained blocked after recovery.

10. Findings and Analyst Interpretation

Finding 1 - Containment was successful: The Windows endpoint could not reach 192.168.92.128 after the outbound firewall rule was applied. Both ICMP and TCP port 80 tests failed after containment.

Finding 2 - Isolation was performed correctly: Ethernet0 was disabled after traffic blocking was validated, creating a second layer of containment while the endpoint was reviewed.

Finding 3 - No visible persistence indicator was identified from the performed checks: The reviewed startup entries and scheduled tasks did not show an obvious unauthorized entry related to the phishing simulation. This is an evidence-based conclusion limited to the checks performed; it is not a claim of a complete disk-level forensic clean bill.

Finding 4 - Recovery was controlled: The endpoint was reconnected to the lab network and regained communication with the default gateway. However, access to the known phishing server remained blocked, which shows that recovery did not undo containment controls.

11. Security Recommendations

Control Area	Recommendation
URL Filtering	Deploy URL filtering or DNS filtering to block known phishing infrastructure and private suspicious links in real time.
Email Security	Use email gateway controls, external sender warnings, attachment/link inspection, and phishing simulation training.
MFA	Enforce multi-factor authentication to reduce the impact of credential exposure.
Browser Hardening	Enable reputation-based protection and block potentially unwanted applications where operationally feasible.
Logging	Continue collecting Sysmon and SIEM telemetry for browser connections to suspicious destinations.
Response Process	Maintain a documented containment checklist covering firewall blocking, isolation, artifact cleanup, persistence checks, and recovery validation.

12. Conclusion

Week 3 was completed successfully for the phishing simulation scenario. The Windows victim endpoint was contained through a targeted firewall rule and hard network isolation, phishing-related browser artifacts were removed, common persistence areas were reviewed, and the system was safely returned to network operation. Final validation confirmed that the endpoint could communicate with the lab gateway while the known GoPhish phishing server remained blocked on port 80.

The response workflow demonstrates a complete and defensible Week 3 process: root cause analysis, containment, eradication checks, recovery, and post-recovery validation. The documentation is aligned with the incident response lifecycle requirement for containment, eradication, and recovery after an evidence-confirmed phishing simulation.